

Hints & Solutions

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Q1 (1) - Single Correct

$$||z|^2 - 2|z|\cos\alpha| \leq |z^2 + 2z\cos\alpha| \leq 1$$

$$\Rightarrow |z|^2 - 2|z| \leq 1 \Rightarrow (|z| - 1)^2 \leq 2$$

$$\Rightarrow |z| \leq \sqrt{2} + 1$$

Hence, maximum value of $|z|$ is $1 + \sqrt{2}$.

Q2 (1) - Single Correct

$$\frac{a}{|z_2 - z_3|} = \frac{b}{|z_3 - z_2|} = \frac{c}{|z_1 - z_2|} = k \text{ [say]}$$

$$a = k|z_2 - z_3|$$

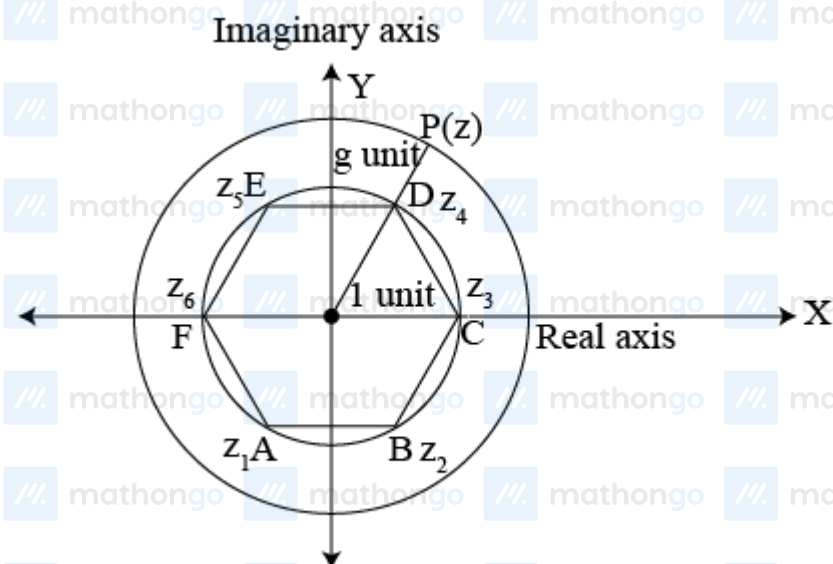
$$\Rightarrow a^2 = k^2(z_2 - z_3)(\bar{z}_2 - \bar{z}_3)$$

$$\Rightarrow \frac{a^2}{(z_2 - z_3)} = k(\bar{z}_2 - \bar{z}_3)$$

$$\text{So, } \frac{a^2}{z_2 - z_3} + \frac{b^2}{z_3 - z_1} + \frac{c^2}{z_1 - z_2}$$

$$\Rightarrow k^2(\bar{z}_2 - \bar{z}_3) + k^2(\bar{z}_3 - \bar{z}_1) + k^2(\bar{z}_1 - \bar{z}_2) = 0$$

Q3 (4) - Single Correct



$$\sum PA^2 = \sum_{i=1}^6 |z - z_i|^2 \text{ Where } z(j = 1, 2, 3, 4, 5, 6) \text{ are roots of } z^6 = 1 \text{ and } |z| = 2.$$

$$= \sum_{i=1}^6 |z|^2 + |z_i|^2 - (z\bar{z}_i + z_i\bar{z})$$

$$= \sum_{i=1}^6 \left[4 + 1 - 2 \operatorname{Re} \left\{ \left(e^{i\left(\frac{2\pi}{d} - \theta\right)} \right) \right\} \right] = 5 \times 6 - 0 = 30$$

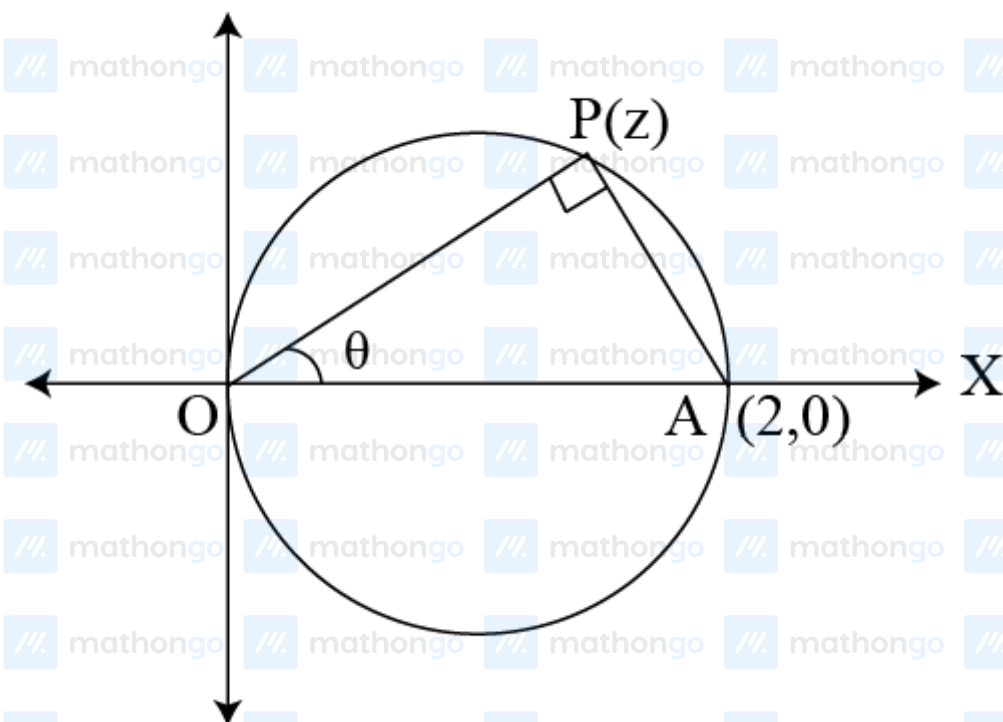
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Q4 (2) - Single Correct

$$\arg\left(\frac{2-z}{0-z}\right) = \frac{\pi}{2}$$



$$\Rightarrow \frac{z-2}{z} = \frac{AP}{OP} i \dots (i)$$

$$\text{and } \tan \theta = \frac{AP}{OP} \dots (ii)$$

From Eqs. (i) and (ii), we get

$$\frac{z-2}{z} = i \tan \theta$$

Q5 (1) - Single Correct

Let $z_1 = x_1$ and $z_2, z_3 = x_2 \pm iy_2$

$$\Rightarrow z_1 + z_2 + z_3 = -\frac{b}{a}$$

$$\Rightarrow x_1 + 2x_2 = -\frac{b}{a} < 0$$

$$\Rightarrow ab > 0 \dots (i)$$

$$\text{Also } z_1 z_2 z_3 = x_1 (x_2^2 + y_2^2) = -\frac{d}{a} < 0$$

$$\Rightarrow ad > 0 \dots (ii)$$

$$\text{Also, } \frac{bc}{a^2} < x_1 (x_2^2 + y_2^2)$$

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$$\Rightarrow \frac{-bc}{a^2} < -\frac{d}{a}$$

$$\therefore bc > ad$$

Q6 (2) - Single Correct

$$\text{Let } A = a + ib$$

$$\Rightarrow |A| = a^2 + b^2$$

$$a^2 + b^2 = 1 \dots (i)$$

$$\text{and } |A + 1| = 1$$

$$\Rightarrow (a + 1)^2 + b^2 = 1 \dots (ii)$$

$$\Rightarrow (a + 1)^2 - a^2 = 0 \text{ [from Eqs. (i) and (ii)]}$$

$$\Rightarrow a^2 + 1 + 2a - a^2 = 0$$

From Eq. (i),

$$a = -\frac{1}{2}$$

$$b^2 = \frac{3}{4} \Rightarrow b = \pm \frac{\sqrt{3}}{2}$$

$$\text{Hence, } A = \frac{-1 \pm i\sqrt{3}}{2} \Rightarrow A = \omega, \omega^2$$

$$\therefore (A)^n = (A + 1)^n$$

$$\Rightarrow \omega^n = (1 + \omega)^n = (-\omega^2)^n$$

Which is true when n must be even or divisible by 3.

For minimum value of n' , n must be 6.

Q7 (3) - Single Correct

$$\prod_{r=1}^{n-1} (\omega - Z_r) = (\omega - z_1)(\omega - z_2)(\omega - z_3) \dots (\omega - z_{n-1})$$

$$= \frac{\omega^n - 1}{\omega - 1}$$

$$\text{When } n = 3k, \text{ then } \prod_{r=1}^{n-1} (\omega - z_r) = 0$$

$$\text{and when } n = 3k + 1, \prod_{r=1}^{n-1} \omega = \frac{\omega^{3k+1} - 1}{\omega - 1} = 1$$

$$\text{Also, when } n = 3k + 2, \prod_{r=1}^{n-1} \omega = \frac{\omega^{3k+2} - 1}{\omega - 1} = \frac{\omega^2 - 1}{\omega - 1} = (\omega + 1)$$

$$\text{Hence, } \prod_{r=1}^{n-1} (\omega - Z_r) \text{ cannot be equal to } -1.$$

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Q8 (2) - Single Correct

Here, $1 + z + z^2 + \dots + z^{50} = (z - z_1)(z - z_2) \dots (z - z_{50})$

Differentiating both sides w.r.t. z after taking logarithm on both sides, we get

$$\frac{1+2z+3z^2+\dots+50z^{49}}{1+z+z^2+\dots+z^{50}} = \frac{1}{z-z_1} + \frac{1}{z-z_2} + \dots + \frac{1}{z-z_{50}}$$

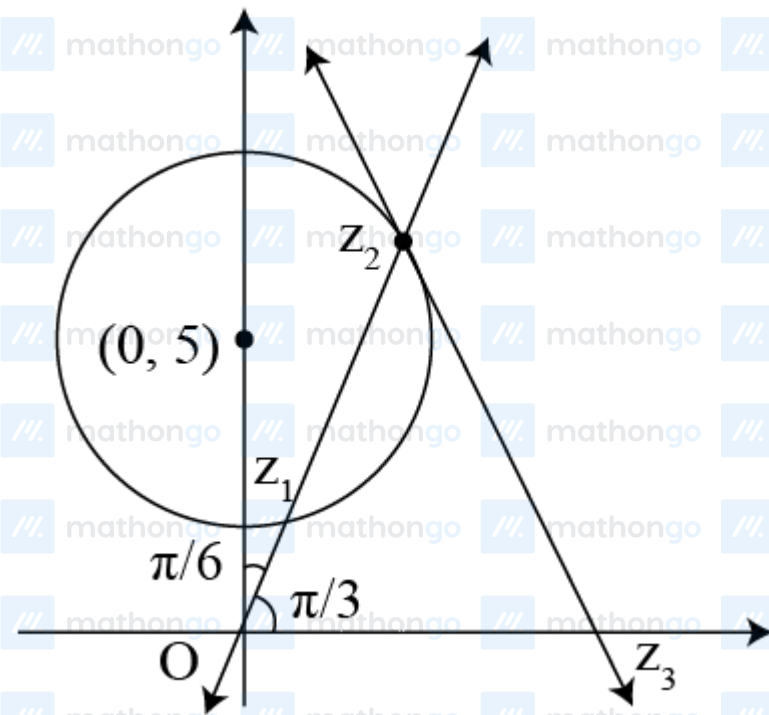
Put $z = 1$

$$\Rightarrow \frac{1+2+3+\dots+50}{1+1+1+\dots+51 \text{ times}} = \frac{1}{1-z_1} + \frac{1}{1-z_2} + \dots + \frac{1}{1-z_{50}}$$

$$\Rightarrow \frac{50(51)}{2 \cdot (51)} = - \left[\frac{1}{z_1-1} + \frac{1}{z_2-1} + \dots + \frac{1}{z_{50}-1} \right]$$

$$\Rightarrow \sum_{r=1}^{50} \frac{1}{z_r-1} = -25$$

Q9 (4) - Single Correct



We have equation of circle,

$$|z - 5i| = 3 \dots (i)$$

Let $z = x + iy$

Then, Eq. (i) of circle,

$$x^2 + (y - 5)^2 = 9 \dots (ii)$$

Also Eqs. (i) and (ii) of line joining z_1 and z_2 is given by

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$$y = \sqrt{3}x \dots (iii)$$

$$[\because \arg(z_1) = \arg(z_2) = \pi/3]$$

Now let us find the intersection of Eqs.(ii) and (iii), we get

$$\frac{y^2}{3} + y^2 + 25 - 10y = 9$$

$$\Rightarrow 4y^2 - 30y + 48 = 0$$

$$\Rightarrow 2y^2 - 15y + 24 = 0$$

$$y = \frac{15 \pm \sqrt{225 - 192}}{4} = \frac{15 \pm \sqrt{33}}{4}$$

$$x = \frac{15 \pm \sqrt{33}}{4\sqrt{3}}$$

Clearly, affix of z_2 is

$$\left(\frac{15 + \sqrt{33}}{4\sqrt{3}}, \frac{15 + \sqrt{33}}{4} \right) \quad [\because |z_1| < |z_2|]$$

Now, equation of tangent to the Eq. (i) of circle is given by

$$xx_1 + yy_1 - 5(y + y_1) + 16 = 0 \dots (iv)$$

where, (x_1, y_1) are affix of z_2

Now, in order to find z_3 , put $y = 0$ in Eq. (iv). we get $xx_1 - 5y_1 + 16 = 0$

$$\Rightarrow x \left(\frac{15 + \sqrt{33}}{4\sqrt{3}} \right) - 5 \left(\frac{15 + \sqrt{33}}{4} \right) + 16 = 0$$

$$\Rightarrow x(15 + \sqrt{33}) - 5\sqrt{3}(15 + \sqrt{33}) + 64\sqrt{3} = 0$$

$$\Rightarrow x = \frac{-64\sqrt{3} + 75\sqrt{3} + 15\sqrt{11}}{15 + \sqrt{33}}$$

$$\Rightarrow x = \frac{11\sqrt{3} + 15\sqrt{11}}{15 + \sqrt{33}}$$

$$\therefore x = \sqrt{11}$$

Hence, $|z_3|$ is $\sqrt{11}$

Q10 (2) - Single Correct

$$z^2 - z + 1 = 0$$

$\therefore z_1$ and z_2 are roots.

$$\therefore z_1^2 - z_1 + 1 = 0$$

$$\text{and } z_2^2 - z_2 + 1 = 0$$

$$\text{Now, } z_1^4 - z_1^3 + 2z_1^2 - 2z_1 + 1$$

$$= z_1^2(z_1^2 - z_1 + 1) + (z_1^2 - z_1 + 1) - z_1 = -z_1$$

$$\text{Similarly, } z_2^4 - z_2^3 + 2z_2^2 - 2z_2 + 1$$

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$$= z_2^2 (z_2^2 - z_2 + 1) + (z_2^2 - z_2 + 1) - z_2$$

$$= -z_2$$

$$\text{Hence } = - \left[\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)^{2013} + \left(\cos \frac{\pi}{3} - i \sin \frac{\pi}{3} \right)^{2013} \right]$$

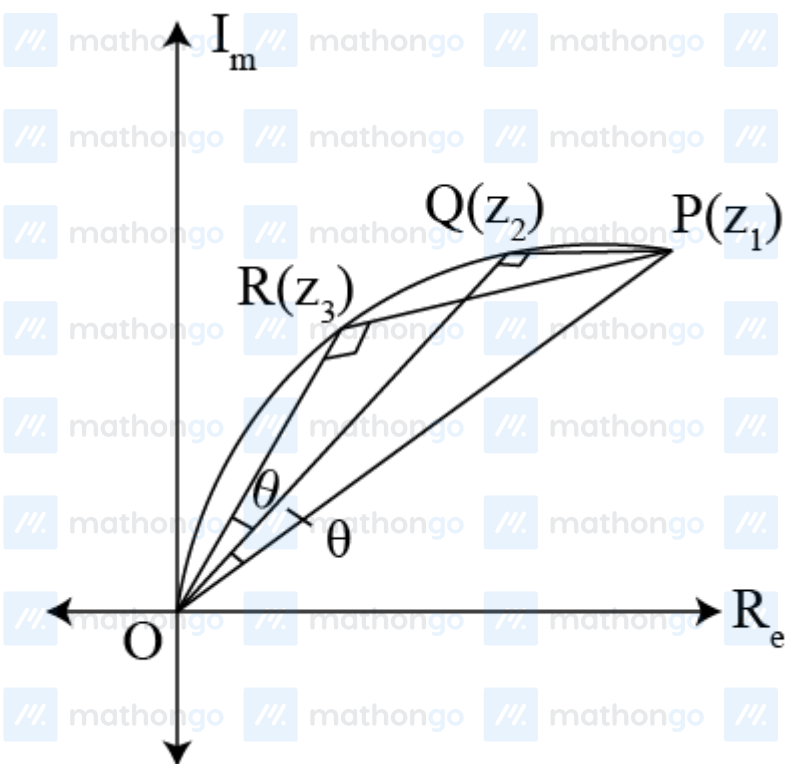
$$= - \left[2 \cos \left(\frac{2013\pi}{3} \right) \right]$$

$$E = 2$$

Q11 (2) - Multiple Correct

$$\cos \theta = \frac{|z_2|}{|z_1|}$$

$$\text{Also, } \cos 2\theta = \frac{|z_3|}{|z_1|}$$



$$\therefore \frac{\cos \theta}{\cos 2\theta} = \frac{|z_2|^2}{|z_1| |z_3|} \quad \dots (i)$$

$$\frac{z_2}{|z_2|} = \frac{z_1}{|z_1|} e^{i\theta} \quad \dots (ii)$$

$$\text{Also, } \frac{z_3}{|z_3|} = \frac{z_2}{|z_2|} e^{i\theta} \quad \dots (iii)$$

On multiplying Eqs. (ii) and (iii), we get

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$$\frac{z_2^2}{|z_2|^2} e^{i\theta} = \frac{z_1 z_3}{|z_1| |z_3|} e^{i\theta}$$

$$\therefore \frac{z_2^2}{z_1 z_3} = \frac{|z_2|^2}{|z_1| |z_3|} \dots (iv)$$

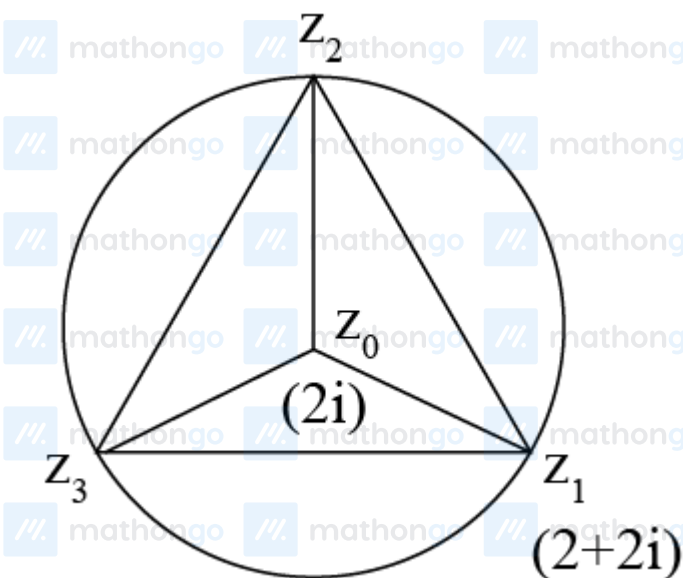
$$\Rightarrow \frac{z_2^2}{z_1 z_3} = \frac{\cos \theta}{\cos 2\theta}$$

Given, $\frac{z_2^2}{z_1 z_3} = \frac{1 + \sqrt{3}}{\sqrt{2}\sqrt{3}} \Rightarrow \frac{1 + \sqrt{3}}{\sqrt{6}} = \frac{\cos \theta}{\cos 2\theta}$

$$\theta = 15^\circ = \pi/12$$

Q12 (1,3) - Multiple Correct

Clearly, the inscribed triangle is equilateral



$$\Rightarrow \frac{z_2 - z_0}{z_1 - z_0} = e^{i2\pi/3}$$

$$\frac{z_3 - z_0}{z_1 - z_0} = e^{-i2\pi/3}$$

$$\Rightarrow z_2 = -1 + i(2 + \sqrt{3})$$

$$\text{and } z_3 = -1 + i(2 - \sqrt{3})$$

Q13 (3,4) - Multiple Correct

$$\left| \frac{z-12}{z-8i} \right| = \frac{5}{3}$$

$$\Rightarrow \left| \frac{(x-12)+iy}{x+i(y-8)} \right| = \frac{5}{3}$$

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$$\Rightarrow \frac{(x-12)^2 + y^2}{x^2 + (y-8)^2} = \frac{25}{9}$$

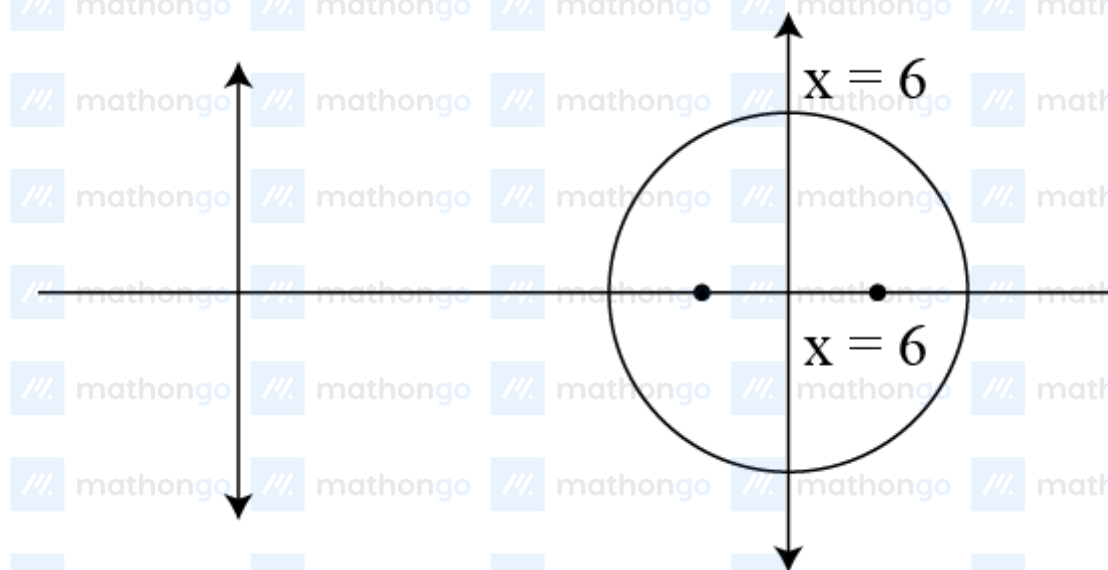
$$\Rightarrow 9(x^2 + 144 - 24x) + 9y^2 = 25x^2 + 25(y^2 + 64 - 16y)$$

$$\Rightarrow 9x^2 + 9 \times 144 - 24 \times 9x + 9y^2$$

$$= 25x^2 + 25y^2 + 25 \times 64 - 16 \times 25y$$

$$\Rightarrow 16x^2 + 16y^2 + 16 \times 19 + 24 \times 9x - 16 \times 25y = 0$$

$$\Rightarrow x^2 + y^2 + \frac{27x}{2} - 25y + 19 = 0 \dots(i)$$



Also, $|z - 4| = |z - 8|$

Perpendicular bisector joining 4 and 8

i.e. $x = 6$

Put $x = 6$ in Eq. (i), we get

$$y = 17, 8$$

Hence, $\text{Im}(z) = 17$ and 8

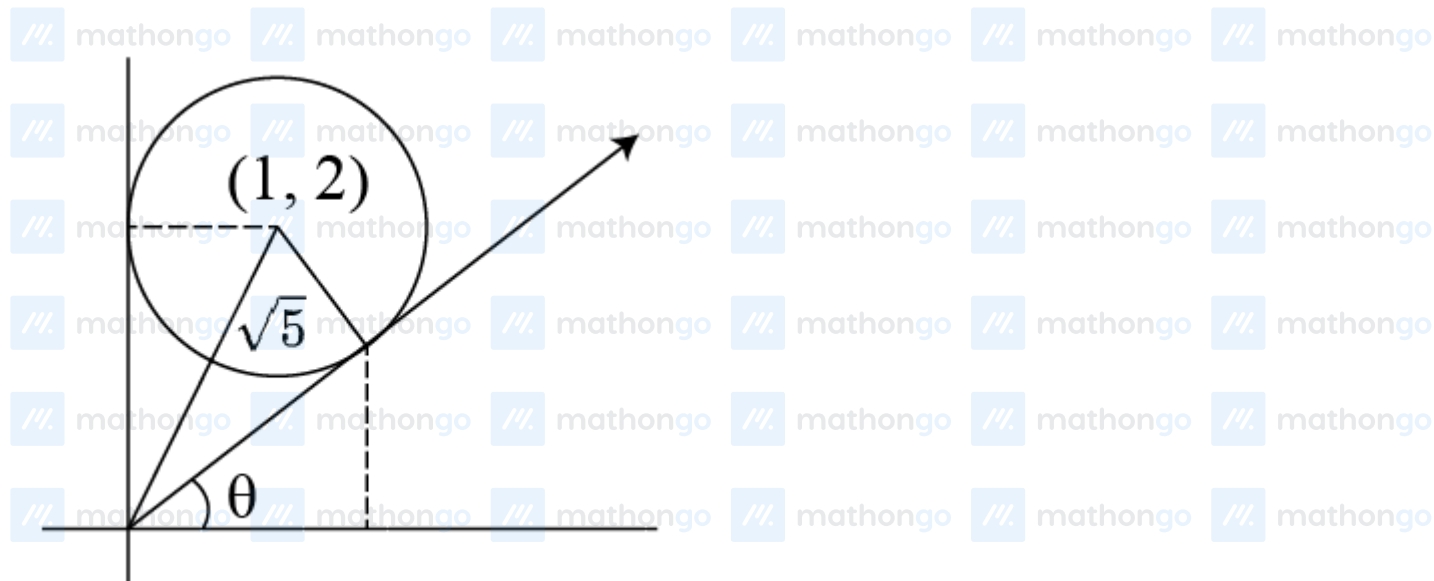
Q14 (1,2,3,4) - Multiple Correct

$$|z - 1 - 2i| \leq 1$$

From figure, maximum argument i.e. $\text{Max}(\arg z) = \frac{\pi}{2}$

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$$\text{Also, } 1 \geq |z - 1 - 2i| = |z - (1 + 2i)|$$

$$1 \geq ||z| - |1 + 2i||$$

$$1 \geq ||z| - \sqrt{5}| \Rightarrow ||z| - \sqrt{5}| \leq 1$$

$$-1 + \sqrt{5} \leq |z| \leq 1 + \sqrt{5}$$

$$\therefore \text{Min } |z| = \sqrt{5} - 1$$

$$\text{and Maximum } (|z|) = \sqrt{5} + 1$$

Q15 (2,3) - Multiple Correct

$$z^2 + z|z| + |z|^2 = 0$$

$$\frac{z^2}{|z|^2} + \frac{z}{|z|} + 1 = 0$$

$$\text{Let } \frac{z}{|z|} = t$$

$$t^2 + t + 1 = 0$$

$$\Rightarrow t = \omega \text{ and } \omega^2$$

$$\text{Hence, } \frac{z}{|z|} = \omega \text{ or } \frac{z}{|z|} = \omega^2$$

$$\Rightarrow z = |z|\omega \text{ or } z = |z|\omega^2$$

$$\Rightarrow z = k\omega \text{ or } z = k\omega^2 \quad [\text{here, } k = |z|]$$

Q16 (1,2) - Multiple Correct

$$\text{Let } z = x + iy$$

$$\text{Given, } \cot^{-1}(\log_3 |2z + 1|) > \cot^{-1}(\log_3 |2z - 1|)$$

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$$\Rightarrow \log_3 |2z + 1| < \log_3 |2z - 1|$$

$$\Rightarrow \log_3 |2z + 1| - \log_3 |2z - 1| < 0$$

$$\Rightarrow \log_3 \left(\frac{|2z+1|}{|2z-1|} \right) < 0$$

$$\Rightarrow \left| \frac{2z+1}{2z-1} \right| < 1$$

$$|2z + 1| < |2z - 1|$$

$$\Rightarrow x < 0$$

Hence, -1 or -2 are possible.

Q17 (1,3) - Multiple Correct

$$|z - 2| = \min\{|z|, |z - 4|\}$$

$$|z - 2| = \begin{cases} |z|, & \text{when } |z| < |z - 4| \\ |z - 4|, & \text{when } |z - 4| < |z| \end{cases}$$

Case I When, $|z| < |z - 4|$, then

$$|z - 2| = |z|$$

Perpendicular bisector of (0,0) and (2,0)

$$\text{i.e. } x = 1$$

Case II When, $|z| > |z - 4|$, then

$$|z - 2| = |z - 4|$$

Perpendicular bisector joining (2,0) and (4,0), i.e. $x = 3$

Hence $\text{Re}(z) = 1$ or 3

Q18 (1,2,3,4) - Multiple Correct

(a) If A, A_2, A_3 are vertices of equilateral triangle.

(b) Then, $(z_1 + \omega z_2 + \omega^3 z_3)(z_1 + \omega^2 z_2 + \omega z_3) = 0$

$$\Rightarrow z_1^2 + z_2^2 + z_3^2 - z_1 z_2 - z_2 z_3 - z_3 z_1 = 0$$

$$(c) \frac{z_2 - z_1}{z_3 - z_2} = \frac{z_3 - z_2}{z_1 - z_3}$$



(d) $\begin{vmatrix} 1 & 1 & 1 \\ z_1 & z_2 & z_3 \\ \bar{z}_2 & \bar{z}_3 & \bar{z}_1 \end{vmatrix} = 0$ are equivalent.

Q19 (1,2,3,4) - Multiple Correct

$$x^n - 1 = (x - 1)(x - \alpha_1)(x - \alpha_2) \cdots (x - \alpha_{n-1})$$

Putting $x = i$, $\frac{i^n - 1}{i - 1} = (i - \alpha_1)(i - \alpha_2) \cdots (i - \alpha_{n-1})$

Case I $i^n = 1$

Then, $(i - \alpha_1)(i - \alpha_2) \cdots (i - \alpha_{n-1}) = 0$

Case II $i^n = -1$, then $(i - \alpha_1)(i - \alpha_2) \cdots (i - \alpha_{n-1}) = \frac{-2}{i-1} = 1 + i$

Case III $i^n = i$ then, $(i - \alpha_1)(i - \alpha_2) \cdots (i - \alpha_{n-1}) = 1$

Case IV $i^n = -i$ then, $(i - \alpha_1)(i - \alpha_2) \cdots (i - \alpha_{n-1}) = \frac{-i-1}{i-1} = i$

Q20 (2,3,4) - Multiple Correct

In option (a) z represents a ellipse and $|z - 2 + i| = \frac{7}{3}$,

$|z - 3| = 2$, $(z - 3 + i)(\bar{z} - 3 - i) = 5$ represents a circle

Q21 (1,2) - Multiple Correct

Given, $|z_1| = |z_2| = |z_3| = 1$

$\Rightarrow z_1 = \frac{1}{z_1}, z_2 = \frac{1}{z_2}, z_3 = \frac{1}{z_3}$

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$$\text{Also, } \frac{z_1^2}{z_2 z_3} + \frac{z_2^2}{z_1 z_3} + \frac{z_3^2}{z_1 z_2} + 1 = 0$$

$$\Rightarrow z_1^3 + z_2^3 + z_3^3 + z_1 z_2 z_3 = 0$$

$$\Rightarrow z_1^3 + z_2^3 + z_3^3 - 3z_1 z_2 z_3 = -4z_1 z_2 z_3$$

$$\Rightarrow (z_1 + z_2 + z_3) \left[(z_1)^2 + (z_2)^2 + (z_3)^2 - \sum z_1 z_2 \right] = -4z_1 z_2 z_3$$

$$\Rightarrow \left(\sum z_1 \right) \left[\left(\sum z_1 \right)^2 - 3 \sum z_1 z_2 \right] = -4z_1 z_2 z_3$$

$$\text{Let } z_1 + z_2 + z_3 = z$$

$$\Rightarrow z \left[z^2 - 3 \sum z_1 z_2 \right] = -4z_1 z_2 z_3$$

$$\Rightarrow z^3 = 3z \cdot \sum z_1 z_2 - 4z_1 z_2 z_3$$

$$\Rightarrow z^3 = z_1 z_2 z_3 \cdot \left[3z \left(\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} \right) - 4 \right]$$

$$= z_1 z_2 z_3 \left[3z (\bar{z}_1 + \bar{z}_2 + \bar{z}_3) - 4 \right]$$

$$\Rightarrow z^3 = z_1 z_2 z_3 \left[3|z|^2 - 4 \right]$$

$$\therefore |z|^3 = |3|z|^2 - 4| \dots (i)$$

$$\text{Now, if } |z| \geq \frac{2}{\sqrt{3}}$$

$$\text{Then, } |z|^3 = 3|z|^2 - 4$$

$$\Rightarrow |z|^3 - 3|z|^2 + 4 = 0$$

$$|z|^2 (|z| - 2) - |z| (|z| - 2) - 2(|z| - 2) = 0.$$

$$\Rightarrow (|z| - 2) (|z|^2 - |z| - 2) = 0$$

$$\Rightarrow (|z| - 2) (|z| - 2) (|z| + 1) = 0$$

$$\therefore |z| = 2 \text{ or } |z| = -1 \quad [\text{rejected}]$$

$$\text{Now, if } 0 < |z| < \frac{2}{\sqrt{3}}, \text{ then Eq. (i) becomes}$$

$$|z|^3 = 4 - 3|z|^2$$

$$\Rightarrow |z|^3 + 3|z|^2 - 4 = 0$$

$$\Rightarrow |z|^2 (|z| + 1) + 4|z| (|z| - 1) + 4(|z| - 1) = 0$$

$$\Rightarrow (|z| + 1) (|z|^2 + 4|z| + 4) = 0$$

$$\Rightarrow (|z| + 1) (|z| + 2)^2 = 0$$

$$\Rightarrow |z| = -1 \text{ or } |z| = -2 [\text{rejected}]$$

$$\therefore |z| = \{1, 2\}, \text{ where } |z| = |z_1 + z_2 + z_3|$$

Hence, options (a) and (b) are correct.

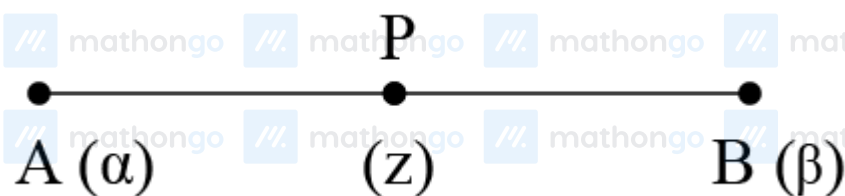
Q22 (1,2,3,4) - Multiple Correct

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$$AP + PB = AB$$



$$\Rightarrow |z - \alpha| + |z - \beta| = |\beta - \alpha|$$

Hence, option (a) is true.

Now,

$$z = \alpha + t(\beta - \alpha)$$

$$= (1 - t)\alpha + (t)\beta, \text{ where } t \in (0, 1)$$

Hence, option (b) is true. Again, $\frac{z - \alpha}{\beta - \alpha}$ is real.

Again, $\frac{z - \alpha}{\beta - \alpha}$ is real.

$$\Rightarrow \frac{z - \alpha}{\beta - \alpha} = \frac{\bar{z} - \bar{\alpha}}{\bar{\beta} - \bar{\alpha}}$$

$$\Rightarrow \begin{vmatrix} z - \alpha & \bar{z} - \bar{\alpha} \\ \beta - \alpha & \bar{\beta} - \bar{\alpha} \end{vmatrix} = 0$$

Hence, option (c) is true.

Also, $\begin{vmatrix} z & \bar{z} & 1 \\ \alpha & \bar{\alpha} & 1 \\ \beta & \bar{\beta} & 1 \end{vmatrix} = 0$ if and only if

$$\begin{vmatrix} z - \alpha & \bar{z} - \bar{\alpha} & 0 \\ \alpha & \bar{\alpha} & 1 \\ \beta - \alpha & \bar{\beta} - \bar{\alpha} & 0 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} z - \alpha & \bar{z} - \bar{\alpha} \\ \beta - \alpha & \bar{\beta} - \bar{\alpha} \end{vmatrix} = 0$$

Hence, option (d) is true.

Q23 (1,2,3) - Multiple Correct

Given, $|z - i| + |z + i| = k, k > 0$

Let $z = x + iy$

$$\Rightarrow |x + |y - i|| + |x + |y + i|| = k$$

$$\Rightarrow |x + i(y - 1)| + |x + i(y + 1)| = k$$

$$\Rightarrow \sqrt{x^2 + (y - 1)^2} + \sqrt{x^2 + (y + 1)^2} = k$$

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$$\Rightarrow \sqrt{x^2 + (y-1)^2} = k - \sqrt{x^2 + (y+1)^2}$$

On squaring both sides, we get

$$\begin{aligned} \left(\sqrt{x^2 + (y-1)^2}\right)^2 &= \left\{k - \sqrt{x^2 + (y+1)^2}\right\}^2 \\ \Rightarrow x^2 + (y-1)^2 &= k^2 + x^2 + (y+1)^2 - 2k\left(\sqrt{x^2 + (y+1)^2}\right) \\ \Rightarrow y^2 + 1 - 2y - y^2 - 1 - 2y - k^2 &= -2k\sqrt{x^2 + (y+1)^2} \\ \Rightarrow -4y - k^2 &= -2k\sqrt{x^2 + (y+1)^2} \\ \Rightarrow 4y + k^2 &= 2k\sqrt{x^2 + (y+1)^2} \end{aligned}$$

Again, on squaring both sides, we get $\Rightarrow 16y^2 + k^4 + 8yk^2 = 4k^2[x^2 + (y+1)^2]$

$$\begin{aligned} \Rightarrow 16y^2 + 8yk^2 + k^4 &= 4k^2(x^2 + y^2 + 1 + 2y) \\ \Rightarrow 16y^2 + k^4 &= 4k^2x^2 + 4k^2y^2 \end{aligned}$$

$$\begin{aligned} \Rightarrow 4k^2x^2 + (4k^2 - 16)y^2 - k^4 &= 0 \\ \Rightarrow 4k^2x^2 + (4k^2 - 16)y^2 &= k^4 \end{aligned}$$

(a) When $k > 2$

let $k = 5$

$$100x^2 + (100 - 16)y^2 = 625$$

$$\Rightarrow 100x^2 + 84y^2 = 625$$

$$\Rightarrow \frac{100x^2}{625} + \frac{84y^2}{625} = 1 \Rightarrow \frac{x^2}{\frac{625}{100}} + \frac{y^2}{\frac{625}{84}} = 1$$

This is equation of ellipse.

$\therefore$ Options (a) and (c) are correct.

(b) When $k = 2$

$$16x^2 + 0 \cdot y^2 = 16 \Rightarrow 16x^2 = 16$$

$$\Rightarrow x^2 = 1$$

$$\therefore x = \pm 1$$

which represent a straight line parallel to Y -axis.

$\therefore$ Option (b) is correct.

Q24 (1,3,4) - Multiple Correct

Given, $||z + i| - |z - i|| = k$

Let $z = x + iy \Rightarrow ||x + iy + i| - |x + iy - i|| = k$

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$$\Rightarrow ||x + i(y + 1)| - |x + i(y - 1)|| = k$$

$$\Rightarrow |\sqrt{x^2 + (y + 1)^2} - \sqrt{x^2 + (y - 1)^2}| = k$$

$$\Rightarrow |\sqrt{x^2 + (y + 1)^2}| - |\sqrt{x^2 + (y - 1)^2}| = k$$

$$\Rightarrow k + |\sqrt{x^2 + (y - 1)^2}| = |\sqrt{x^2 + (y + 1)^2}|$$

On squaring both sides, we get

$$k^2 + x^2 + (y - 1)^2 + 2k\sqrt{x^2 + (y - 1)^2} = x^2 + (y + 1)^2$$

$$\Rightarrow k^2 + x^2 + y^2 + 1 - 2y + 2k\sqrt{x^2 + (y - 1)^2}$$

$$= x^2 + y^2 + 1 + 2y$$

$$\Rightarrow k^2 - 4y + 2k\sqrt{x^2 + (y - 1)^2} = 0$$

$$\Rightarrow k^2 - 4y = -2k\sqrt{x^2 + (y - 1)^2}$$

Again, on squaring both sides, we get

$$(k^2 - 4y)^2 = 4k^2(x^2 + y^2 + 1 - 2y)$$

$$\Rightarrow k^4 + 16y^2 - 8k^2y = 4k^2x^2 + 4k^2y^2 - 8k^2y + 4k^2$$

$$\Rightarrow (16 - 4k^2)y^2 - 4k^2x^2 - 4k^2 + k^4 = 0$$

$$\Rightarrow 4k^2x^2 - (16 - 4k^2)y^2 + 4k^2 - k^4 = 0 \dots (i)$$

(a) When k lies between 0 and 2 i.e. $0 < k < 2$.

Put $k = 1$ in Eq. (i), we get

$$\Rightarrow 4x^2 - \frac{4x^2}{3} = 1 \Rightarrow \frac{y^2}{1/4} - \frac{x^2}{(3/4)} = 1$$

which is an equation of hyperbola.

Hence, option (a) is correct.

(b) When $k > 2$ Put $k = 3$ in Eq. (i), we get

$$36x^2 + 20y^2 - 45 = 0$$

$$\Rightarrow \frac{x^2}{\frac{45}{36}} + \frac{y^2}{\frac{45}{20}} = 1$$

which represent equation of ellipse.

Hence, option (b) is incorrect.

(c) When $k = 0 \Rightarrow -16y^2 = 0$

$$\Rightarrow 16y^2 = 0 \Rightarrow y^2 = 0$$

Hints & Solutions

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$$\therefore y = 0$$

It represent a straight line. Hence, option (c) is correct.

$$\Rightarrow 16x^2 = 0 \Rightarrow x^2 = 0$$

$$\therefore x = 0$$

which is Y -axis. which is origin.

Hence, option (d) is correct.

Q25 (1,2) - Multiple Correct

Clearly, P and Q are on the opposite sides of the origin such that $OP = OQ$

Therefore, $OP = OQ$ and $OQ = OPe^{i\pi}$

$$\Rightarrow |a + ib| = |c + id|$$

$$\text{and } c + id = (a + ib)e^{i\pi}$$

$$\Rightarrow |a + ib| = |c + id|$$

$$\text{and } c + id = (a + ib)(\cos \pi + i \sin \pi)$$

$$\Rightarrow |a + ib| = |c + id|$$

$$\text{and } c + id = (a + ib)(-1 + 0 \cdot i)$$

$$\Rightarrow |a + ib| = |c + id|$$

$$\text{and } c + id = -a - ib$$

$$\Rightarrow |a + ib| = |c + id|$$

$$\text{and } c = -a, d = -b$$

$$\Rightarrow |a + ib| = |c + id| \text{ and } a + c = 0$$

$$b + d = 0$$

$$\Rightarrow |a + ib| = |c + id| \text{ and } a + c = b + d$$

Q26 (3) - Comprehension Type

Given, z_1 and z_2 are two complex numbers satisfying

$$3z_1^2 - 2z_1z_2 + 2z_2^2 = 0 \dots(i)$$

$$\text{and } \operatorname{Re}\left(\frac{z-2}{z+2}\right) = 0 \dots(ii)$$

$$\therefore \operatorname{Re}\left(\frac{z-2}{z+2}\right) = 0$$

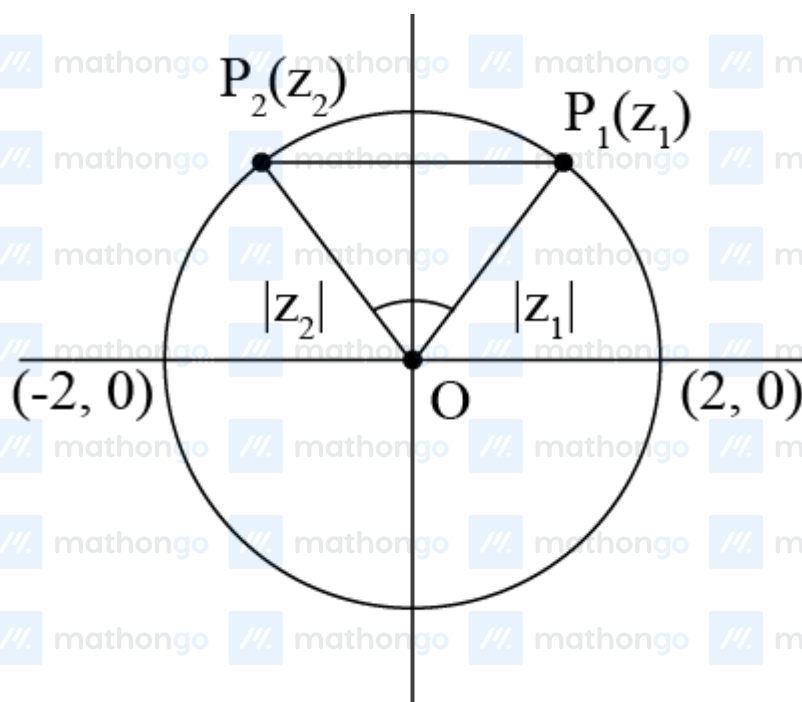
$\therefore \frac{z-2}{z+2}$ is purely imaginary.

$$\Rightarrow \arg\left(\frac{z-2}{z+2}\right) = \frac{\pi}{2}$$

Hints & Solutions

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$\Rightarrow z$ is a circle whose end of diameter are $(2,0)$ and $(-2,0)$ ii)



$\because P_1$ and P_2 satisfies both the Eqs. (i) and (ii), we get

$$|z_1| = OP_1 = 2$$

Q27 (1) - Comprehension Type

$$3z_1^2 - 2z_1z_2 + 2z_2^2 = 0 \Rightarrow 3t^2 - 2t + 2 = 0$$

Putting $\frac{z_1}{z_2} = t$

$$\therefore t = \frac{2 \pm \sqrt{4-24}}{2} = 1 \pm i\sqrt{5} \Rightarrow \frac{z_1}{z_2} = 1 + i\sqrt{5}$$

$$\Rightarrow \arg\left(\frac{z_1}{z_2}\right) = \sqrt{5}$$

$$\therefore \sin \theta = \frac{\sqrt{5}}{\sqrt{6}}$$

$$\begin{aligned} \text{Area of } \triangle OP_1P_2 &= \frac{1}{2} |z_1| |z_2| \sin \theta = \frac{1}{2} |z_1| \cdot \frac{\sqrt{6}}{2} |z_2| \cdot \sin \theta \\ &= \frac{1}{2} \cdot 2 \cdot \frac{\sqrt{6}}{2} \cdot 2 \cdot \sqrt{\frac{5}{6}} = \sqrt{5} \end{aligned}$$

Q28 (2) - Comprehension Type

$$z_3 + i\omega z_2 = z_1 + i\omega z_1 \Rightarrow z_3 - z_1 = i\omega (z_1 - z_2)$$

$$\Rightarrow \frac{z_3 - z_1}{z_2 - z_1} = -i\omega \Rightarrow \frac{z_3 - z_1}{z_2 - z_1} = e^{i\frac{\pi}{6}}$$

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$$\Rightarrow |z_3 - z_1| = |z_2 - z_1|$$

$$\therefore \triangle ABC \text{ is isosceles and } \angle A = \frac{\pi}{6}$$

Q29 (4) - Comprehension Type

$$G = \frac{z_1 + z_2 + z_3}{3}$$

$$\therefore \frac{z_3 - z_0}{z_2 - z_0} = e^{i\frac{\pi}{3}} = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} = -\omega^2 = 1 + \omega$$

$$\Rightarrow z_3 - z_0 = (1 + \omega)(z_2 - z_0) = z_2 - z_0 + \omega z_2 - \omega z_0$$

$$\Rightarrow z_3 - z_2 - \omega z_2 = -\omega z_0$$

$$\Rightarrow \omega z_0 = (1 + \omega)z_2 - z_3$$

$$\frac{z_1 + z_2 + z_3}{3} \text{ or the centre}$$

3



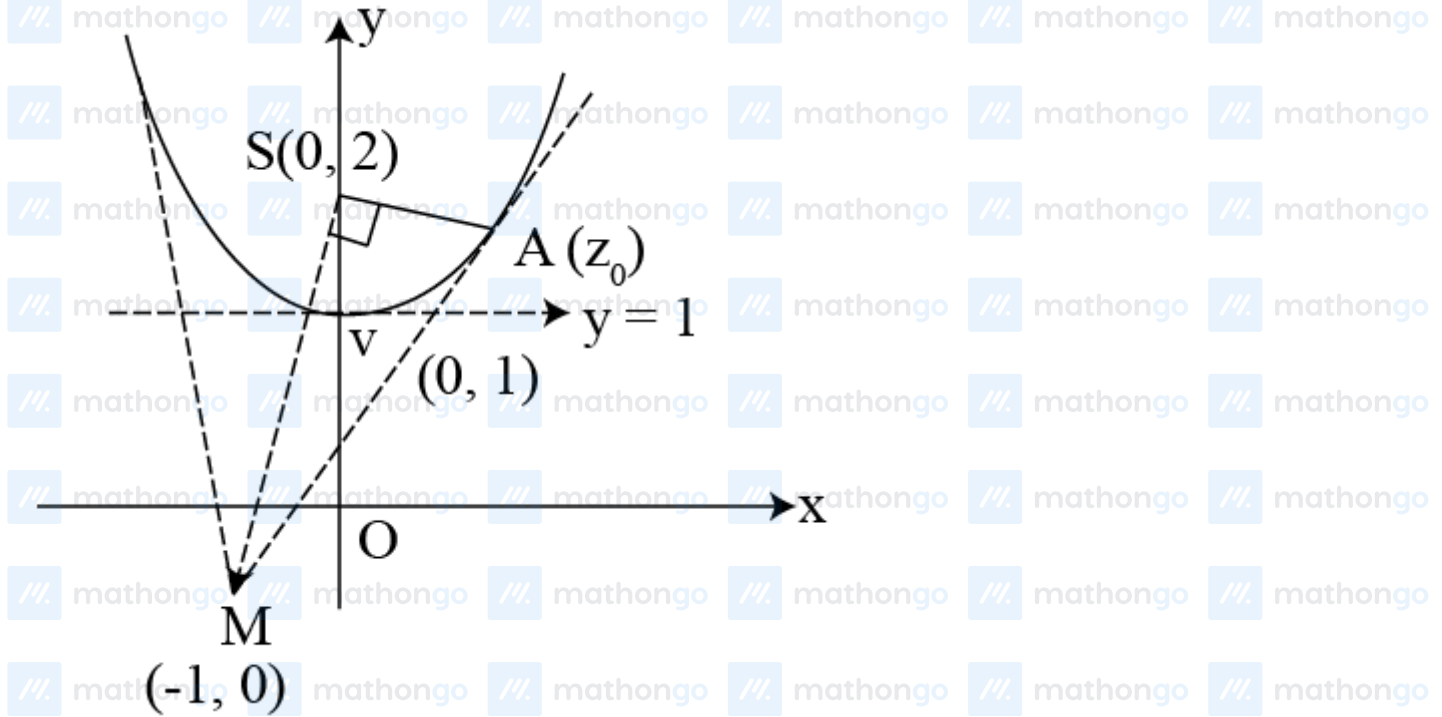
1 : 2

$$z_0 = \frac{(1 + \omega)z_2 - z_3}{\omega}$$

Q30 (1) - Comprehension Type

Hints & Solutions

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From above figure, using property of parabola, $\angle ASM = 90^\circ$

By, using rotation, we get $\arg\left(\frac{2i-z_0}{2i+1}\right) = \frac{\pi}{2}$

So, $\frac{2i-z_0}{2i+1}$ is purely imaginary.

$$\therefore \frac{2i-z_0}{2i+1} + \frac{\bar{z}_0+2i}{2i-1} = 0$$

Q31 (4) - Comprehension Type

Using property of parabola, the equation of circle whose ends of diameter are z_0 and $2i$ is

$$\left| z - \left(\frac{z_0+2i}{2} \right) \right| = \frac{|z_0-2i|}{2}, \text{ which always touches the tangent at vertex, which is } y = 1, \text{ so } \text{Im}(z) \geq 1$$

Q32 (4) - Comprehension Type

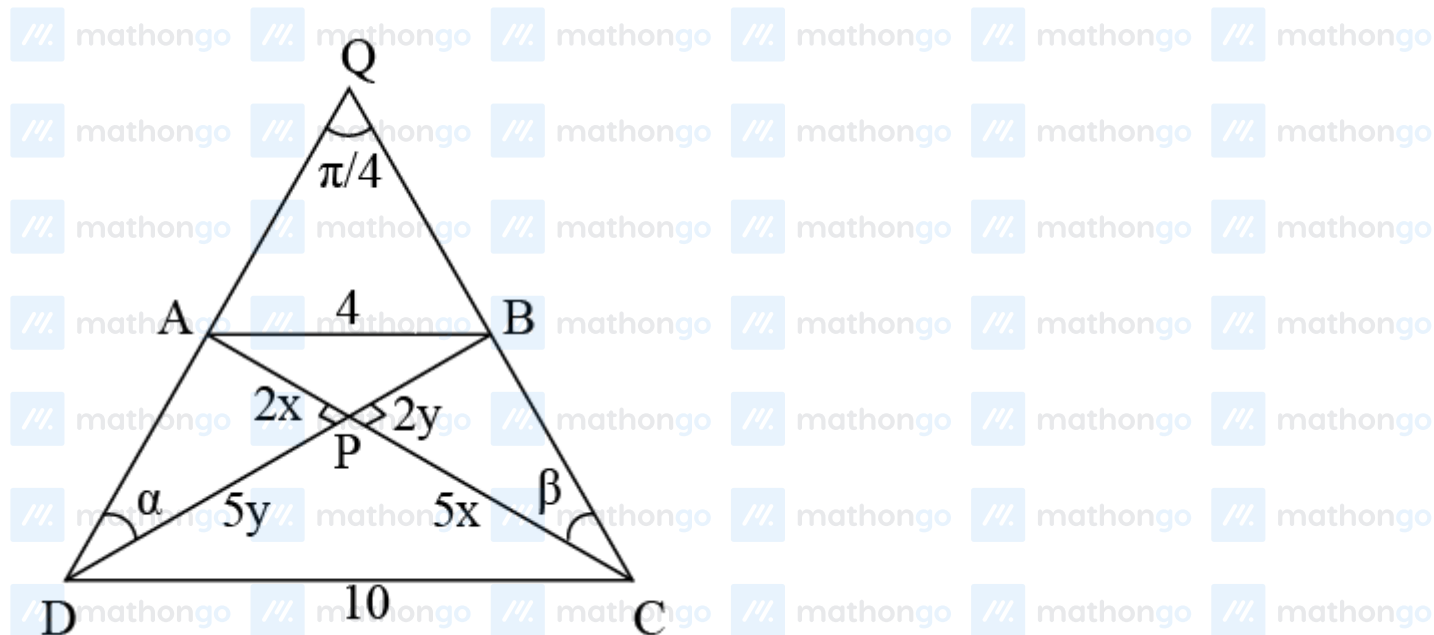
Solution (For Question 32, 33)

Here, $\triangle ABP$ and $\triangle CDP$ are similar Let $AP = 2x, BP = 2y$, then $CP = 5x, DP = 5$

$$\therefore \text{Area of trapezium } ABCD = \frac{49}{2}xy$$

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$$\text{Also, } \tan \alpha = \frac{2x}{5y}, \tan \beta = \frac{2y}{5x}$$

$$\alpha + \beta = 45^\circ$$

$$\Rightarrow \tan(\alpha + \beta) = 1$$

$$\Rightarrow \frac{\frac{2x}{5y} + \frac{2y}{5x}}{1 - \frac{2x}{5y} \cdot \frac{2y}{5x}} = 1$$

$$\Rightarrow xy = \frac{10}{21}(x^2 + y^2)$$

$$\text{Also, } AB^2 = AP^2 + BP^2$$

$$\Rightarrow x^2 + y^2 = 4$$

$$\therefore x = \frac{40}{21}$$

$$\Rightarrow \text{Area of } \triangle PCB = \frac{1}{2}(2y)(5x) = 5xy$$

$$\text{Area of trapezium } ABCD = \frac{49}{2} \cdot \frac{10}{21}(x^2 + y^2)$$

$$= \frac{7 \times 5}{3} \times 4 = \frac{140}{3}$$

Q33 (2) - Comprehension Type

$$\text{Area of } \triangle PCB = 5(xy) = 5 \times \frac{10}{21}(x^2 + y^2)$$

$$= \frac{50}{21} \cdot 4 = \frac{200}{21}$$

Q34 (2) - Comprehension Type

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Solution (For Question 34, 35)

For A, $|z + 1| \leq 2 + \operatorname{Re}(z)$

$$\Rightarrow (x + 1)^2 + y^2 \leq 4 + 4x + x^2$$

$$\Rightarrow y^2 \leq 3 + 2x$$

$$\Rightarrow y^2 \leq 2\left(x + \frac{3}{2}\right)$$

For B $|z - 1| \geq 1$

$$(x - 1)^2 + y^2 \geq 1$$

For C, $|z - 1|^2 \geq |z + 1|^2$

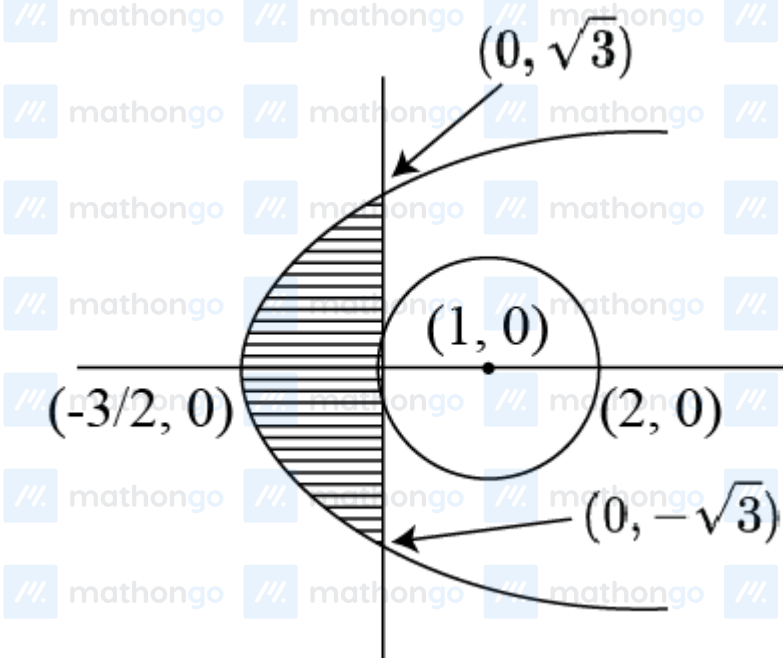
$$\Rightarrow (z - 1)(\bar{z} - 1) \geq (z + 1)(\bar{z} + 1)$$

$$\Rightarrow z\bar{z} - z - \bar{z} + 1 \geq z\bar{z} + z + \bar{z} + 1$$

$$\Rightarrow z + \bar{z} \leq 0$$

$$\therefore x \leq 0$$

$(-1, 0), (-1, 1), (-1, -1), (0, 0), (0, 1), (0, -1)$, but $z = -1$ is not in the domain in set C.



Hence, total number of point(s) having integral coordinates in the region $A \cap B \cap C$ is 5.

Hints & Solutions

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Q35 (1) - Comprehension Type

$$\text{Required area} = 2 \int_{-3/2}^0 \sqrt{2 \left(x + \frac{3}{2} \right)} dx = 2\sqrt{3} \text{sq units}$$

Q36 (1) - Matrix Match (Only one option is correct)

$$(i) |x|^2 = x \cdot \bar{x}$$

$$= (z_1 + z_2 + z_3)(\bar{z}_1 + \bar{z}_2 + \bar{z}_3)$$

$$= \sum |z_1|^2 + \sum (z_2 + z_3) \bar{z}_1$$

Similarly, we can find $|y|^2$ and $|z|^2$.

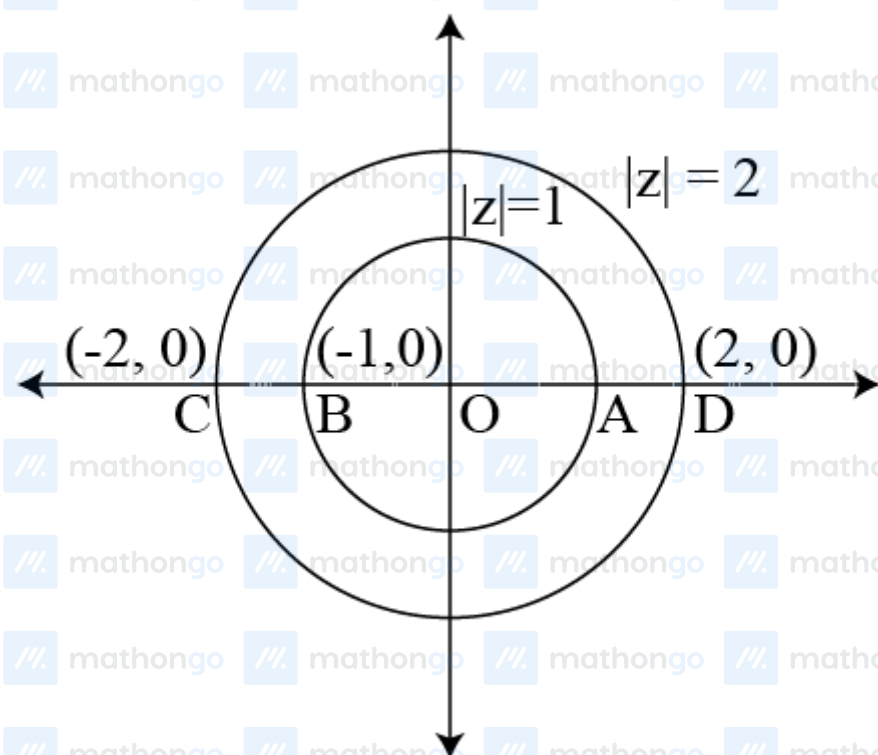
$$(iv) \sum |x|^2 = |x|^2 + |y|^2 + |z|^2$$

$$= 3 \left\{ |z_1|^2 + |z_2|^2 + |z_3|^2 \right\} + \sum \bar{z}_1 \left\{ z_2 (1 + \omega + \omega^2) + z_3 (1 + \omega + \omega^2) \right\}$$

$$= 3 \sum |z_1|^2$$

Hence (i) → (s), (ii) → (p), (iii) → (q), (iv) → (r)

Q37 (2) - Matrix Match (Only one option is correct)



From figure it is clear that

$$(i) \text{ Maximum value of } |z_1 + z_2| = AC = AB + BC = 3$$

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(ii) Minimum value of $|z_1 - z_2| = AD = OD - OA = 2 - 1 = 1$

(iii) Minimum value of $|2z_1 - 3z_2| \geq |2|z_1| - 3|z_2|| = 4$

(iv) Maximum value of $|z_1 + 2z_2| \leq |z_1| + 2|z_2| \leq 1 + 4 = 5$

$\therefore \text{Max}|z_1 + 2z_2| = 5$

Hence, (i) $\rightarrow$ (p), (ii) $\rightarrow$ (q), (iii) $\rightarrow$ (r), (iv) $\rightarrow$ (s)

Q38 (1) - Matrix Match (Only one option is correct)

$z^2 - z + 1 = 0 \Rightarrow z = -\omega, -\omega^2$

$\Rightarrow \arg(z) = \frac{-\pi}{3}, \frac{\pi}{3}$

$\therefore$ (i) $\rightarrow$ (q, r)

(ii) $z^2 + z + 1 = 0$

$z = \omega, \omega^2 \Rightarrow \arg(z) = \frac{2\pi}{3} \text{ or } -\frac{2\pi}{3}$

$\therefore$ (ii) $\rightarrow$ (p, s)

(iii) $2z^2 + 1 + i\sqrt{3} = 0$

$z^2 = -\frac{1}{2} - i\frac{\sqrt{3}}{2} = \cos \frac{-2\pi}{3} + i \sin \left(\frac{-2\pi}{3} \right)$

$\arg(z) = \frac{2\pi}{3} \text{ or } -\frac{\pi}{3}$

$\therefore$ (iii) $\rightarrow$ (q, s)

(iv) $z^2 = -\frac{1}{2} + \frac{i\sqrt{3}}{2} = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$

$\arg(z) = \frac{\pi}{3} \text{ or } -\frac{2\pi}{3}$

$\therefore$ (iv) $\rightarrow$ (p, r)

Hence, (i) $\rightarrow$ q, r, (ii) $\rightarrow$ p, s, (iii) $\rightarrow$ q, s, (iv) $\rightarrow$ p, r

Q39 (2) - Integer Type

Here, the factors of $x^2 - x + 1$ are $-\omega, -\omega^2$.

So, $x^{n+1} - x^n + 1$ is divisible by $x^2 - x + 1$

$\therefore (-\omega)^{n+1} - (-\omega)^n + 1 = 0$

$\Rightarrow (-1)^{n+1} \cdot (\omega^{n+1} + \omega^n) + 1 = 0$

$\Rightarrow (-1)^{n+1} \omega^n (-\omega^2) + 1 = 0$

Hints & Solutions

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$$\Rightarrow -(-1)^{n+1} \cdot \omega^{n+2} + 1 = 0$$

Then, $n + 2$ is single of 3 and n is odd.

$$\Rightarrow n = 6m + 1$$

$$\therefore n(s) = 6, n(E) = 1$$

$$\Rightarrow P = \frac{1}{6}$$

$$\therefore 12P = 2$$

Q40 (1) - Integer Type

Given, $z + w = i$... (i)

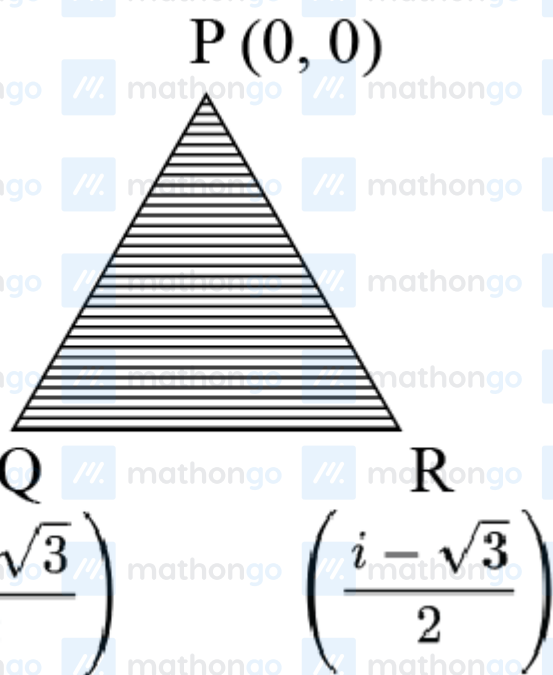
and $z^2 + w^2 = 1$... (ii)

On squaring both sides of Eq. (i), we get

$$z^2 + w^2 + 2zw = -1$$

$$\Rightarrow 1 + 2zW = -1$$

$$\Rightarrow ZW = -1$$



Now let us consider a quadratic equation in x whose roots are z and w . $x^2 - ix - 1 = 0$

$$x^2 - ix - 1 = 0$$

$$x = \frac{i \pm \sqrt{i^2 + 4}}{2} = \frac{i \pm \sqrt{3}}{2}$$

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$$\therefore A = \ar(\Delta PQR) = \frac{1}{2} \begin{vmatrix} 0 & 1 \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{vmatrix} = \frac{1}{2} \left(\frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4} \right) = \frac{\sqrt{3}}{4}$$

$$\therefore 80A^2 = 80 \times \frac{3}{16} = 15 = 15 \times 10^0 = 1.5 \times 10^1$$

$$\therefore x = 1$$

Q41 (6) - Integer Type

Note that, $z = 0$ is one root of the given equation,

$$z^4 + az^3 + (12 + 9i)z^2 + bz = 0$$

Cancelling z , we get

$$z^3 + 9z^2 + (12 + 9i)z + b = 0$$

The 3 vertices of the square other than $0 + 0i$ are u , iu , $u(1 + i)$

$$\text{Now } u(ui) + iu(1 + i)u + [(1 + i)u] \cdot u = 12 + 9i$$

$$\Rightarrow u^2[i + i - 1 + 1 + i] = 12 + 9i$$

$$\Rightarrow 3iu^2 = 12 + 9i$$

$$\Rightarrow u^2 = \frac{4+3i}{i} = 3 - 4i$$

$$u = \sqrt{3 - 4i} = \pm(2 - i)$$

$$\therefore u = 2 - i \text{ or } -2 + i$$

Again,

$$-a = u + ui + u(1 + i)$$

$$= u(1 + i + 1 + i) = 2u(1 + i)$$

$$= 2(2 - i)(1 + i) = 2(3 + i) = 6 + 2i$$

$$\therefore a = -6 - 2i$$

$$\text{Again, } u(ui) \cdot (1 + i)u = -b$$

$$\Rightarrow -b = u^3(-1 + i)$$

$$\Rightarrow b = (2 - i)^3(1 - i) = (2 - 11i)(1 - i)$$

$$= 2 - 2i - 11i - 11 = -9 - 13i$$

$$\therefore b = -9 - 13i$$

$$\text{Hence, } 2b - 13a = 2(-9 - 13i) - 13(-6 - 2i)$$

Hints & Solutions

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$$= 60 = 6 \times 10$$

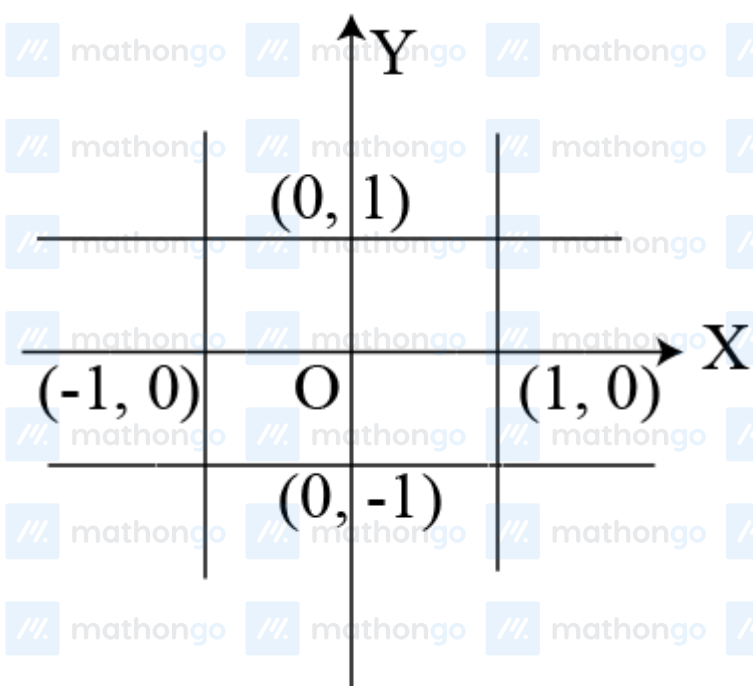
$$\therefore x = 6$$

Q42 (2) - Integer Type

Clearly, S denotes a region bounded by $x = \pm 1$ and $y = \pm 1$. i.e. square of side 2 on the complex plane.

Hence, $n(S) = 4 \dots (i)$

$$\text{Now, let } w = \frac{3}{4}(1+i)(x+iy) = \frac{3}{4}[(x-y) + i(x+y)]$$

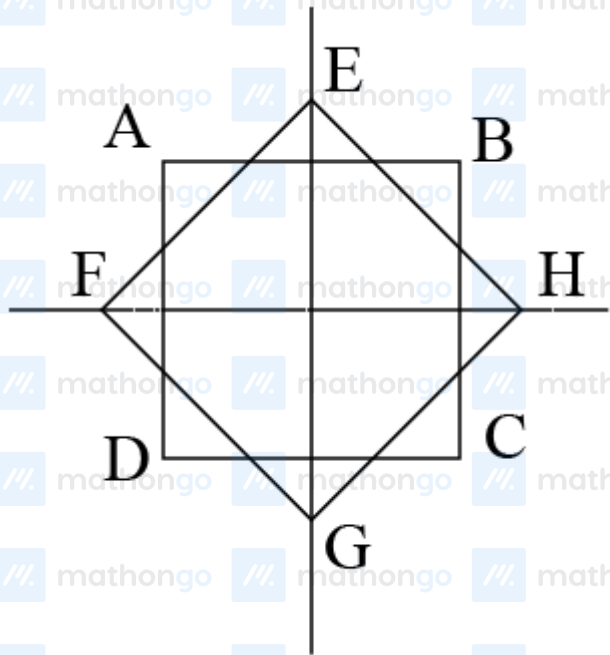


In order that w lies on S .

$$-1 \leq \operatorname{Re}(w) \leq 1 \text{ and } -1 \leq \operatorname{Im}(z) \leq 1$$

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Hence, $-1 \leq \frac{3}{4}(x - y) \leq 1$

and $-1 \leq \frac{3}{4}(x + y) \leq 1$

$\Rightarrow -\frac{4}{3} \leq (x - y) \leq \frac{4}{3}$

and $-\frac{4}{3} \leq (x + y) \leq \frac{4}{3}$

Area of EFGH = $2 \cdot \left(\frac{4}{3}\right) \cdot 2 \cdot \left(\frac{4}{3}\right) \cdot \frac{1}{2} = \frac{32}{9} = \frac{m}{n}$

$\therefore m + n = 32 + 9 = 41$

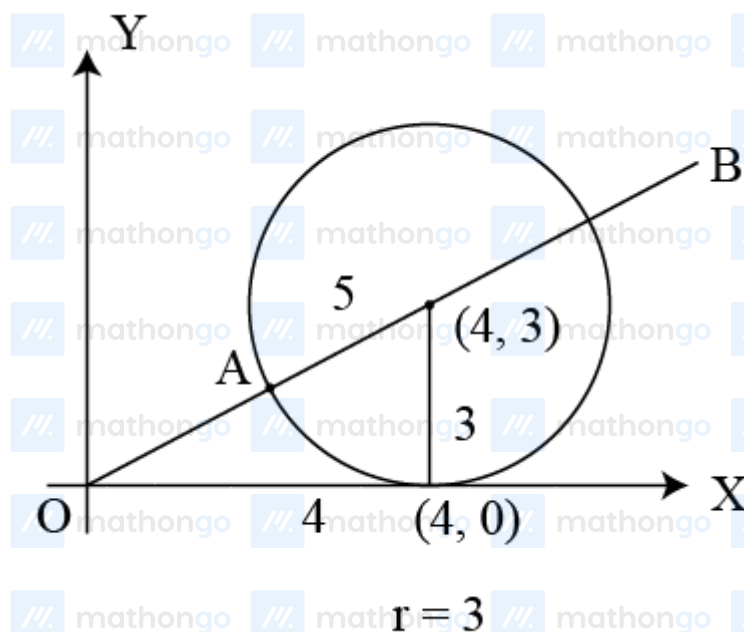
Number of digits in 41 is 2.

Q43 (8) - Integer Type

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Since, z has only real value which belongs to m .



$$\therefore r = 3$$

Minimum value of $|z| = OB = 5 + 3 = 8$

Q44 (4) - Integer Type

$$\text{Given, } |2z + 3i| = |z|^2$$

$$|z|^2 \leq 2|z| + 3 \Rightarrow |z|^2 - 2|z| - 3 \leq 0$$

$$(|z| + 3)(|z| - 1) \leq 0$$

$$\Rightarrow |z|^2 \leq 2|z| + 3 \Rightarrow |z|^2 - 2|z| - 3 \leq 0 \Rightarrow (|z| - 3)(|z| + 1) \leq 0$$

$$-1 \leq |z| \leq 3$$

$$\therefore m = 1, M = 3$$

$$\text{Hence, } M + m = 3 + 1 = 4$$

Q45 (2) - Integer Type

The given equality can be written as $z_1^3 + z_2^3 + z_3^3 + z_1 z_2 z_3 = 0$

$$\text{Or } z_1^3 + z_2^3 + z_3^3 - 3z_1 z_2 z_3 = -4z_1 z_2 z_3$$

$$\Rightarrow (z_1 + z_2 + z_3)(z_1^2 + z_2^2 + z_3^2 - z_1 z_2 - z_2 z_3 - z_3 z_1) = -4z_1 z_2 z_3 \dots (i)$$

$$\text{Let } z_1 + z_2 + z_3 = z$$

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$$\therefore (z_1 + z_2 + z_3) \left((z_3 + z_2 + z_1)^2 - 3(z_1z_2 + z_2z_3 + z_3z_1) \right)$$

$$= -4z_1z_2z_3$$

$$\Rightarrow z(z^2 - 3(z_1z_2 + z_2z_3 + z_3z_1)) = -4z_1z_2z_3$$

$$\Rightarrow z^3 = 3z(z_1z_2 + z_2z_3 + z_3z_1) - 4z_1z_2z_3$$

$$= z_1z_2z_3 \left[3z \left(\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} \right) - 4 \right]$$

$$= z_1z_2z_3 (3z(\bar{z}_1 + \bar{z}_2 + \bar{z}_3) - 4) = z_1z_2z_3 (3|z|^2 - 4)$$

Taking modulus on both sides, we get

$$|z|^3 = |3|z|^2 - 4| \dots (ii)$$

If $|z| \geq \frac{2}{\sqrt{3}}$, then

$$\Rightarrow |z|^3 - 3|z|^2 + 4 = 0$$

$$\Rightarrow |z| = 2$$

If $|z| \leq \frac{2}{\sqrt{3}}$, then

$$\Rightarrow |z|^3 + 3|z|^2 - 4 = 0$$

$$|z| = 1$$

Hence, the possible values of $|z_1 + z_2 + z_3|$ is 1 or 2.

Q46 (1) - Integer Type

$$\text{Given, } w = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$$

$$\Rightarrow \begin{vmatrix} z+1 & w & w^2 \\ w & z+w^2 & 1 \\ w^2 & 1 & z+w \end{vmatrix} = 0$$

Applying $R_1 \rightarrow R_1 + R_2 + R_3$, we get

$$\begin{vmatrix} z & z & z \\ w & z+w^2 & 1 \\ w^2 & 1 & z+w \end{vmatrix} = 0$$

$$\Rightarrow z(z^2) = 0 \Rightarrow z^3 = 0$$

$$\therefore z = 0$$

Hence $y = 0$ is only one value of z .

Q47 (3) - Integer Type

Hints & Solutions

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Let $w = e^{i2\pi/3} = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$.

Then, $\frac{|x|^2 + |y|^2 + |z|^2}{|a|^2 + |b|^2 + |c|^2}$

$$= \frac{[|a+b+c|^2 + |a+bw+cw^2|^2 + |a+bw^2+cw|^2]}{3(|a|^2 + |b|^2 + |c|^2)}$$

$$= \frac{3(|a|^2 + |b|^2 + |c|^2)}{|a|^2 + |b|^2 + |c|^2} = 3$$

Q48 (2) - Integer Type

Minimum value of $|1+z| + |1-z|$

Now, $|1+z| + |1-z| \geq |1+z+1-z|$

$|1+z| + |1-z| \geq |2|$

$|1+z| + |1-z| \geq 2$

Hence, minimum value of $|1+z| + |1-z|$ is 2.

Q49 (1) - Integer Type

Given system of equation is $|z+1-i| = 2 \dots(i)$

and $\text{Re}(z) \geq 1 \dots(ii)$

Put $z = x+iy$ in Eq. (i), we get

$|x+iy+1-i| = 2$

$|(x+1)+i(y-1)| = 2$

$\Rightarrow \sqrt{(x+1)^2 + (y-1)^2} = 2$

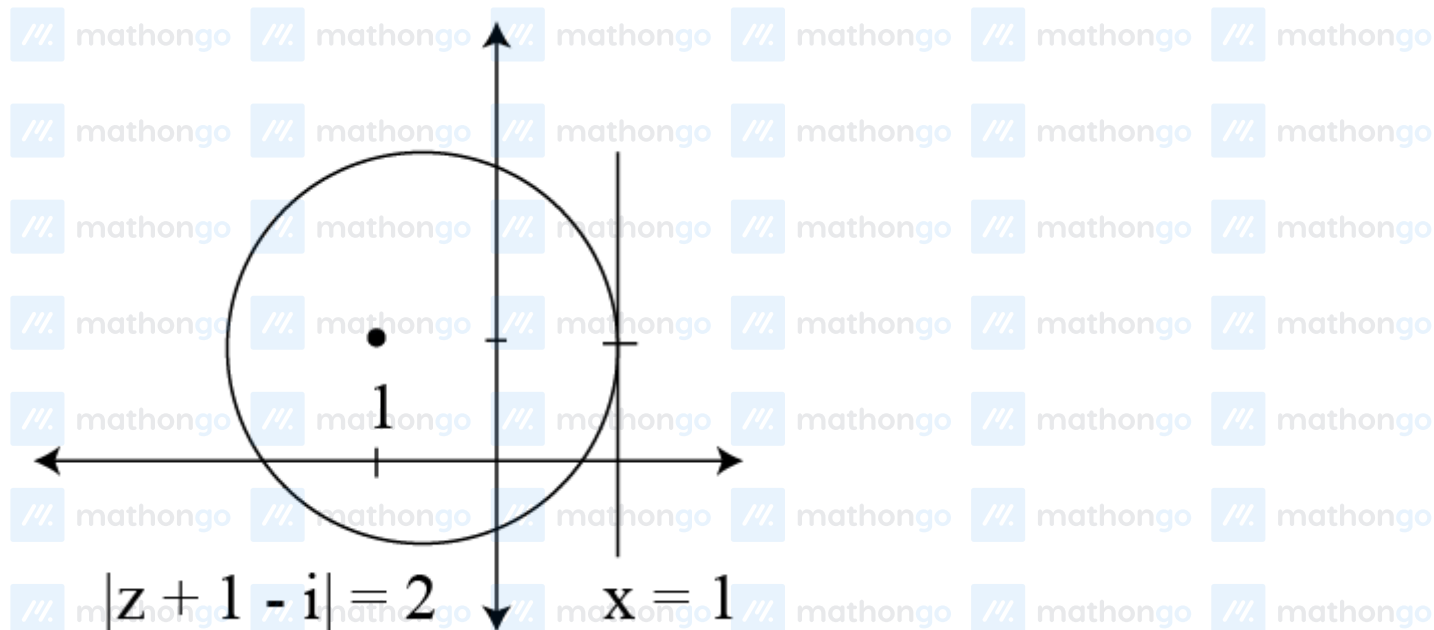
$\Rightarrow (x+1)^2 + (y-1)^2 = (2)^2 \dots(iii)$

which represent a circle having centre $(-1,1)$ and radius 2 and $\text{Re}(z) = x \geq 1$

Represent all points on the right of the line $x = 1$ in argand plane. The graph of Eqs. (iii) and (iv) is given alongside. Since, two curves touch each other at only one point.

Hints & Solutions

#MathBoleTohMathonGo



Hence, only one solution is possible.

Q50 (2) - Integer Type

The minimum value of $|z - 1|^2 + |z - 2|^2 + |z - 3|^2$

Put $z = 1$, we get $0 + 1 + 4 = 5$

Put $z = 2$, we get $1 + 0 + 1 = 2$

Put $z = 3$, we get, $4 + 1 + 0 = 5$

Hence the minimum value is 2 and maximum value is 5.

$\therefore$ Minimum value of $|z - 1|^2 + |z - 2|^2 + |z - 3|^2$ is 2.